

## Project Design Phase

### Proposed Solution

|              |                                                     |
|--------------|-----------------------------------------------------|
| Date         | 22 September 2026                                   |
| Project Name | Online Complaint Registration and Management System |

#### Proposed Solution Template:

| S.No. | Parameter                                | Description                                                                                                                                                                                                                                                      |
|-------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Problem Statement (Problem to be solved) | Citizens and customers lack a transparent, centralized way to report issues, and organizations lack an efficient system to assign, track, and resolve those complaints securely at scale.                                                                        |
| 2.    | Idea / Solution description              | A full-stack MERN platform (React frontend, Express/Node.js backend, MongoDB persistence) where users register, submit complaints with category and attachments, and track status in real time, while admins assign complaints to agents and monitor resolution. |
| 3.    | Novelty / Uniqueness                     | Role-based JWT authentication distinguishing User, Agent, and Admin access; a defined complaint status lifecycle (OPEN → IN_PROGRESS → RESOLVED → CLOSED); and automatic complaint closure triggered by user feedback submission.                                |
| 4.    | Social Impact / Customer Satisfaction    | Gives citizens real-time visibility into complaint status and accountability (assigned agent, progress timeline), reducing frustration from opaque phone- or email-based complaint channels and building trust in the resolution process.                        |
| 5.    | Business Model (Revenue Model)           | Deployable as an internal grievance-redressal tool for organizations or municipal bodies (per-department licensing/subscription), or offered as a white-labeled SaaS complaint-management platform for businesses.                                               |
| 6.    | Scalability of the Solution              | Stateless JWT authentication and a decoupled REST API allow horizontal scaling of the Node.js backend, while MongoDB's document model supports growing complaint volumes and additional agent/department collections without schema rework.                      |